Appendix C
Differential Equations

The same form of simple differential equations is used to model very different problems throughout this text. They are presented here along with their solutions. The solutions can be checked by substituting them in the differential equation and showing that the equation is satisfied. This appendix is not meant to serve as a primer on differential equations or their solutions.

Solutions to first- and second-order differential equations, respectively, have one and two free parameters that are satisfied by the conditions of the problem. When the independent variable is time, t, these conditions are called *initial conditions*, so for the dependent variable $q(t)$, q is specified at a given time, such as at $t = 0$ for a first-order differential equation. For a second-order equation, both q and $\mathrm{d}q/\mathrm{d}t$ at $t = 0$ can be given. The dependent variable q can be a coordinate, such as x and angle θ or something else, such as force F. When the independent variable is a spatial coordinate, such as x, these conditions are called *boundary conditions*.

Unless otherwise specified, F and G are constants.

C.1 Simple First- and Second-Order Differential Equations

In these differential equations the derivatives of the dependent variable, q, depend on the independent variable t.

First-Order, Constant Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}q}{\mathrm{d}t} = F \tag{C.1}$$

has solution

$$q(t) = Ft + q(t = 0), \tag{C.2}$$

where $q(0) = q(t = 0)$ is the initial condition.

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This type of equation is used to describe the temperature rise of the body with metabolic heating and no heat loss (6.36).

Second-Order, Constant Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} = F \tag{C.3}$$

has solution

$$q(t) = \frac{Ft^2}{2} + \frac{\mathrm{d}q(0)}{\mathrm{d}t}\, t + q(0), \tag{C.4}$$

where the initials conditions are q and $\mathrm{d}q/\mathrm{d}t$ evaluated at $t = 0$.

This type of equation is used in the model of ball throwing (3.80).

Second-Order, Increasing Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} = F + Gt \tag{C.5}$$

has solution

$$q(t) = \frac{Ft^2}{2} + \frac{Gt^3}{6} + \frac{\mathrm{d}q(0)}{\mathrm{d}t}\, t + q(0), \tag{C.6}$$

where the initials conditions are q and $\mathrm{d}q/\mathrm{d}t$ evaluated at $t = 0$.

This type of equation is used in the model for bending a cantilever (4.44) with the position x as the independent variable.

First- and Second-Order, Increasing Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}(t\ \mathrm{d}q/\mathrm{d}t)}{\mathrm{d}t} = Gt \tag{C.7}$$

or equivalently

$$t\,\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \frac{\mathrm{d}q}{\mathrm{d}t} = Gt \tag{C.8}$$

has solution

$$q(t) = q(0) + \frac{Gt^2}{4}, \tag{C.9}$$

where the initial condition for q is evaluated at $t = 0$, and q and $\mathrm{d}q/\mathrm{d}t$ are finite at $t = 0$.

This type of equation is used in determining the viscous flow in a tube (7.31).

C.2 Exponential Decay and Drag

In these differential equations the first derivative of the dependent variable, q, depends on q and in some cases on the dependent variable t.

First-Order, Proportional Drag, No Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{\tau} = 0 \tag{C.10}$$

decays in time t as

$$q(t) = q(0)\exp(-t/\tau), \tag{C.11}$$

where $q(0)$ is the initial condition and τ has units of time (s) and is called a *time constant*. The value of q decays exponentially in time with this characteristic time constant. This can be due to a "frictional force" or *damping* with a rate $1/\tau$.

This type of equation is used in the mechanical models of non-Hookean materials ((4.22) with the strain ϵ as the independent variable) to describe the speed when there is Stokes-type drag that is proportional to speed (7.72) and in describing pulsatile flow in (8.100), (8.111), and Problem 8.62. This equation is equivalent to the first two terms of (C.30) describing position.

This type of equation is used in the viscoelastic mechanical models of materials with constant driving terms, including the Maxwell (4.52), Voigt (4.57), and Kelvin/standard linear (4.68) models, a model of muscles (5.6), the arterial pulse (Problem 8.62), and temperature regulation (13.18).

Equation (C.10) can also be phrased as

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \gamma q = 0, \tag{C.12}$$

where the damping constant $\gamma = 1/\tau$ is defined. The solution (C.11) becomes

$$q(t) = q(0)\exp(-\gamma t). \tag{C.13}$$

First-Order, Proportional Drag, Constant Driving

A variation of (C.10),

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{\tau} = F \tag{C.14}$$

with constant term F, has solution:

$$q(t) = (q(0) - F\tau)\exp(-t/\tau) + F\tau. \tag{C.15}$$

This type of equation is used in the viscoelastic mechanical models of materials with constant driving terms, including the Maxwell (4.52), Voigt (4.57), and Kelvin/standard linear (4.68) models, a model of muscles (5.6), the arterial pulse (Problem 8.62), and temperature regulation (13.18).

First-Order, Proportional Drag, Increasing Driving

A variation of (C.14) includes a driving term that varies linearly with the independent variable

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{\tau} = F + Gt. \tag{C.16}$$

It has solution

$$q(t) = (F\tau - G\tau^2)(1 - \exp(-t/\tau)) + q(0)\exp(-t/\tau) + Gt\tau. \tag{C.17}$$

This type of equation is used in the Kelvin/standard linear viscoelastic mechanical model with a linearly increasing driving term (4.72).

First-Order, Proportional Drag, Arbitrary Temporal Driving

A variation of (C.12) and (C.14) includes a driving term that varies arbitrarily on the independent variable

$$\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{\tau} = F(t). \tag{C.18}$$

Substituting $q(t) = s(t)\exp(-t/\tau)$ into this gives

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \exp(t/\tau)\ F(t), \tag{C.19}$$

so

$$s(t) = s(0) + \int_0^t \exp(t'/\tau)F(t')\mathrm{d}t' \tag{C.20}$$

and

$$q(t) = \exp(-(t/\tau))\left(q(0) + \int_0^t \exp(t'/\tau)F(t')\mathrm{d}t'\right). \tag{C.21}$$

This type of equation is used for pulsatile flow (8.109).

First-Order, Higher-Order Drag, No Driving

A variable $q(t)$ obeying

$$\frac{\mathrm{d}q}{\mathrm{d}t} + Aq^n = 0 \tag{C.22}$$

varies as

$$q(t) = \left(q(0)^{1-n} + (n-1)At\right)^{1/(1-n)}, \tag{C.23}$$

for $n \neq 1$, where $q(0)$ is the initial condition. For $n = 1$, see (C.10) and (C.11).

For $n = -2$ this describes flow with resistance and compliance (8.16)

For $n = 2$ this describes the equation of motion for hydrodynamic drag where q is speed (7.77). Then

$$q(t) = \frac{q(0)}{1 + Aq(0)t}, \tag{C.24}$$

If $q = \mathrm{d}p/\mathrm{d}t$, where p would be the position for this type of drag, then

$$p(t) = p(0) + \frac{1}{A}\ \ln(1 + Aq(0)t). \tag{C.25}$$

C.3 Harmonic Oscillator

In these differential equations, the second derivative of the dependent variable, q, depends on q and in some cases on the dependent variable t.

Harmonic Oscillator: Undamped, Not Driven

A variable $q(t)$ obeying

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \omega_0^2 q = 0 \tag{C.26}$$

oscillates as

$$q(t) = A\cos(\omega_0 t + \phi), \tag{C.27}$$

where A is the amplitude, ω_0 is the resonant frequency of this harmonic oscillator (with units rad/s), and ϕ is the phase. Alternatively, this solution can be expressed as

$$q(t) = B\cos(\omega_0 t) + C\sin(\omega_0 t), \tag{C.28}$$

where B and C are amplitudes. The frequency, f, is $\omega/2\pi$, and has units of Hz (Hertz) or cps (cycles per second), and so (C.27) would be

$$q(t) = A\cos(2\pi f_0 t + \phi). \tag{C.29}$$

This type of equation is used in the models of the harmonic motion of a mass on a spring (3.7), the simple (3.14) and complex pendulums (3.26), and Euler buckling (4.86).

Harmonic Oscillator: Damped, Not Driven

Adding damping to the harmonic oscillator equation (C.26) with damping constant $\gamma = 1/\tau$ gives

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \gamma \frac{\mathrm{d}q}{\mathrm{d}t} + \omega_0^2 q = 0, \tag{C.30}$$

with solution

$$q(t) = A \exp(-\gamma t/2) \cos(\omega_0 t + \phi), \tag{C.31}$$

where A is the amplitude, ω_0 is the resonant frequency of this harmonic oscillator (with units rad/s), and ϕ is the phase. This solution is not exact, but is valid for $\omega_0 \gg \gamma$. This harmonic oscillation damps in a time $\sim 1/\gamma$, which corresponds to about $\omega_0/(2\pi\gamma)$ cycles; ω_0/γ is often called the quality factor Q of the system, as is discussed in the Chap. 10 discussion of acoustic resonances and more generally in Appendix D.

This type of equation is used in the models of harmonic oscillators, and simple and complex pendulums.

Harmonic Oscillator: Undamped, Driven

The equation

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \omega_0^2 q = F \cos(\omega t) \tag{C.32}$$

looks like the equation of motion for a simple harmonic oscillator of frequency ω_0 (C.26) with an extra term (the last one), which drives the oscillator with a "force" that oscillates at a frequency ω; ω can differ from the resonant frequency ω_0. The particular solution to this equation is

$$q(t) = \frac{F}{\omega_0^2 - \omega^2} \cos(\omega t), \tag{C.33}$$

to which the solution (C.27), $q(t) = A \cos(\omega_0 t + \phi)$ (of the homogeneous equation (C.26)) is added to set the initial conditions by the proper choice of A and ϕ. Without the driving term ($F = 0$), the solution is the usual harmonic solution (C.27).

This type of equation is used in the models of pulsatile blood flow (8.47) and the general models in Appendix D (D.2).

Harmonic Oscillator: Damped, Driven

If ω_0 were to approach ω, the response for the undamped, driven harmonic oscillator, (C.33), would approach infinity because of this resonance. There is always some damping that adds a term $\gamma\, \mathrm{d}q/\mathrm{d}t$ to (C.32) to give the new equation of motion

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \gamma \frac{\mathrm{d}q}{\mathrm{d}t} + \omega_0^2 q = F \cos(\omega t). \tag{C.34}$$

This has a particular and steady-state solution

$$q(t) = \frac{\left(\omega_0^2 - \omega^2\right) F}{\left(\omega_0^2 - \omega^2\right)^2 + (\gamma\omega)^2} \cos(\omega t). \tag{C.35}$$

The homogeneous solution (C.31), $q(t) = A \exp(-\gamma t/2) \cos(\omega_0 t + \phi)$ for $\omega_0 \gg \gamma$, is added to this to set the initial conditions by the proper choice of A and ϕ. Without the driving term ($F = 0$), the solution is the usual damped harmonic solution (C.31).

This type of equation is used in the models of pulsatile blood flow (8.49), acoustic impedance (10.21), and the general models in Appendix D (D.2).

C.4 Partial Differential Equations

Partial differential equations contain derivatives of more than one independent variable.

The Diffusion Equation

The diffusion equation (7.66) in one dimension (x) has the form

$$D \frac{\partial^2 q}{\partial x^2} = \frac{\partial q}{\partial t}. \tag{C.36}$$

The formal solution gives $q(x, t)$ from $q(x', t = 0)$, the distribution for all x (called x') at an earlier time (defined as $t = 0$). It is

$$q(x, t) = \frac{Q}{\sqrt{4\pi D t}} \int_{-\infty}^{\infty} q(x', 0) \exp\left(-(x - x')^2/4Dt\right) \mathrm{d}x', \tag{C.37}$$

where Q is the integral of q over all x at any time—which means that the total amount of the entity undergoing diffusion, such as the mass or number of particles, does not change during diffusion.

The importance of this diffusion is most simply seen when the initial distribution is gaussian and has an initial spread $\sigma(0)$,

$$q(x, 0) = \frac{Q}{\sqrt{2\pi\sigma^2(0)}} \exp\left(-x^2/2\sigma^2(0)\right). \tag{C.38}$$

Then the solution becomes

$$q(x, t) = \frac{Q}{\sqrt{2\pi\sigma^2(t)}} \exp\left(-x^2/2\sigma^2(t)\right), \tag{C.39}$$

where

$$\sigma^2(t) = \sigma^2(0) + 2Dt. \tag{C.40}$$

If the initial spread is not gaussian, the solution is slightly different but approaches this for large x and/or large t. (Sometimes σ is defined a bit differently than it is here, as in (7.68).)

This type of equation is used in diffusion (7.66).

The integral over a gaussian probability curve,

$$\text{erf}(x) = \frac{2}{\pi^{1/2}} \int_0^x \exp(-z^2)\text{d}z, \tag{C.41}$$

is known as the error function. It increases from 0 to 1 as x increases from 0 to ∞. The error function is used in the statistics describing head injury as in Fig. 3.63.

The Poisson–Boltzmann Equation

The Poisson–Boltzmann Equation (12.44) is of the form

$$\nabla^2 q = \kappa^2 q. \tag{C.42}$$

where ∇^2 is the Laplacian. In one dimension $\nabla^2 q = \partial^2 q/\partial x^2$, while in Cartesian coordinates in three dimensions it is $\nabla^2 q = \partial^2 q/\partial x^2 + \partial^2 q/\partial y^2 + \partial^2 q/\partial z^2$. In three dimensions it can be expressed as $\nabla^2 q = (1/r)(\text{d}^2(rq)/\text{d}r^2)$ when there is no angular dependence (spherical symmetry), where r is the radial coordinate. Using this in (C.42) gives

$$\frac{1}{r}\frac{\text{d}^2(rq)}{\text{d}r^2} = \kappa^2 q. \tag{C.43}$$

Replacing $rq(r)$ by $p(r)$, this reduces to

$$\frac{\text{d}^2 p}{\text{d}r^2} = \kappa^2 p, \tag{C.44}$$

with solution $p(r) = p(0)\exp(-\kappa r)$ valid for all r, and so

$$q(r) = q(0)\frac{\exp(-\kappa r)}{r}. \tag{C.45}$$

This is used to determine the potential of a charge in a neutral region with mobile charges, as in (12.44) and Problem 12.8.